StudySearch Zoho API v2 — Developer Guide

zoho-api-v2 · Issue 2 · 19 October 2026 · Pilot

For	The Zoho developers: the Zoho Creator counsellor app (course search) and the Zoho CRM sync (Institutes and Courses modules)
API	One HTTPS address for Zoho API v2, fifteen actions
Environment	Pilot (live). Production will have its own address and keys, issued at go-live
Keys	Two keys, one for Zoho Creator and one for the Zoho CRM sync, sent separately by a secure channel. They are never written in this guide or the files that come with it
Files with this guide	`StudySearch-Zoho-API-v2.openapi.json` (OpenAPI 3.1) and `StudySearch-Zoho-API-v2.postman_collection.json` (Postman; set the variable `apiKey`)
Replaces	Issue 1 of this guide (16 October 2026), and the Zoho Developer Handover v1.0 (1 September 2026) for new work. The v1 address keeps working until you have moved (section 11)
New in issue 2	Section 5 now sets out the two ways to give Zoho Creator course search: live search through the API, or a copy held in Creator (seeded from CSV, then kept current with `sync`), and which to use when. English test code DET (Duolingo). Counts measured on 19 October 2026

1. What the API gives you

Courses, universities and colleges (providers), campuses, provider contacts, scholarships, rankings, statistics (QILT and PRISMS) and the vocabularies for your filters, for Australia, New Zealand and Canada. For the CRM it also gives a change feed and CSV files keyed by CRICOS code, Australia first.

Five rules apply to every response:

1. **Null means not held.** A value the platform does not hold is `null` or an empty list. Nothing is filled with a default or an estimate.
2. **Only published data is returned.** A provider the platform has switched off is not returned, nor are its courses, campuses, contacts and scholarships. Every course reads `publication_status: "published"`.
3. **One address per version.** This guide covers `zoho-api-v2` only. Within v2, fields, filters and actions are only added, never renamed or removed: ignore fields you do not use. A breaking change would come as `zoho-api-v3` at a new address, with its own guide, and v2 would keep running alongside it.
4. **CRICOS codes are the keys for Australia.** Every Australian provider and course carries `cricos_provider_code` and `cricos_course_code`. Every one of the 1,546 Australian providers and 26,103 Australian courses has one, and no course code is shared.
5. **Third-party material carries attribution.** Rankings and logos come with `attribution` and `terms_status` (approved for display). Show the attribution with them.

2. Quick start

Zoho Creator (counsellor search)

1. Put the Creator key in a Zoho Connection (custom service, API key in the `Authorization` header as `Bearer <key>`). Never in Deluge source or widget JavaScript.
2. Call `reference` once a day and cache the vocabularies (section 7).
3. Choose how search works (section 5.1). Recommended: search live with `search` (section 5.2); keep a copy of Institutes and Courses in Creator for lookups, proposals and reports (section 5.3).
4. Call `course` or `courses` when a counsellor opens a course or a lead lists CRICOS codes (or read them from the copy).

Zoho CRM (Institutes and Courses kept current)

1. Put the CRM key in its own Zoho Connection.
2. Seed once from the CSV files (`export`, section 6.1), matching on CRICOS code, and keep the sync cursor that comes with the first page.
3. Then run `sync` on a schedule (every 15–60 minutes, section 6.2) and apply what changed.

3. Address and keys

```
POST https://fxcwkweaxjtknorudmwp.supabase.co/functions/v1/zoho-api-v2
Content-Type: application/json
Authorization: Bearer <your key>
```

Key	For	Limit
Zoho Creator	Live search and detail from the counsellor app	120 calls a minute
Zoho CRM sync	`export` and `sync` for the CRM modules (it may call every action)	120 calls a minute

- `x-cf-token: <your key>` is accepted instead of the `Authorization` header.
- There is no `contract` field: the address is the version.
- Each key's calls are counted and shown to StudySearch (calls, errors, times). Tell us if a key may have leaked: we switch it off and issue a new one.
- Do not use or ask for Supabase keys, database credentials or table access. This API is the supported way in.

4. Requests and responses

Every request is a POST with a JSON body. `action` chooses what you get. Optional fields can be left out.

```
{ "action": "search", "filters": { "keyword": "nursing", "country_codes": ["AU"] },
"page": 1, "page_size": 20 }
```

Every JSON answer carries `api_version: "zoho-v2"` and a `request_id`. Lists return `total`, `page`, `page_size` and `items`; details return `item`. `filters_not_applied` lists filter keys that were not recognised (check spelling).

HTTP	`error.code`	What to do
200	—	Use the result. A search with no results is 200 with `items: []`
400	`INVALID_JSON`, `INVALID_ACTION`, `INVALID_INPUT`	Fix the request; `error.detail` says what is wrong. Do not retry it unchanged
401	`AUTHENTICATION_REQUIRED`	Check the Connection's key and header
404	`NOT_FOUND`	The course, provider or scholarship

HTTP	error.code	What to do
		is not held or not published
429	RATE_LIMITED	Wait the seconds in Retry-After, then retry
503	SERVICE_UNAVAILABLE	Retry later with a back-off; show a short message to the user

Quote the request_id when you report a problem.

5. Zoho Creator: course search

5.1 Two ways to search, and which to use

Zoho Creator can take StudySearch data the same way the website does: seed a copy once, then fetch only what changed. Course search can then run in one of two ways.

	A. Live search (recommended for the search screen)	B. Search a copy held in Creator
How it works	The search screen sends the counsellor's filters to search and shows the page that comes back	Courses and Institutes are stored as Creator records (seeded from CSV, kept current with sync); the search screen filters those records
Matching	StudySearch relevance: words in the title, provider name and course codes, ranked; English score and band per test, intake months and tuition worked out on our side	Creator's own filters: exact or "contains" matches on fields, no ranking by relevance
English and intakes	Every test (IELTS, PTE, TOEFL iBT, CAE, DET), overall and lowest band	Only the columns you store (the CSV holds IELTS overall and band, PTE overall, intake months)
Freshness	Always current	As current as the last sync run (every 15–60 minutes)
Calls	One call per search (cache 15 minutes)	No call per search; one scheduled sync run every 15–60 minutes
Storage in Creator	None	About 34,400 records for Australia (1,546 Institutes, 26,103 Courses, 3,957 Campuses, 2,794 Contacts); check your plan's record allowance
Works when StudySearch is unreachable	No (show a short message)	Yes, from the copy

Recommendation: use both, as the website does. Search live (A) on the counsellor's search screen, because it gives the full English, intake and relevance logic with no extra build. Keep a copy (B) of Institutes and Courses for everything that is not a search: opening a course, building a student proposal, linking a lead to a course, and Creator reports. If a fully offline search is required, B works with the limits in the table.

5.2 Live search

This answers the two questions in your email of 14 October.

1. Details in the search results. Each search item has the course's intakes (month and, where held, year and start date), English requirements by test (overall and lowest band), tuition (amount, year, basis, annual amount), campuses (city, state, postcode, metro area), study level, study area, duration, delivery modes, the official course link and the scholarship count. Filtering by English score, band and intake month happens on our side, so the page you get back is already filtered: there is no need to call course for each result.

2. The provider in the search. Send the provider name as text in `filters.provider_name` (part of the name is enough: "Queensland" finds The University of Queensland and others). If you have the CRICOS provider code, send `filters.provider_cricos_codes` instead; it is exact. `keyword` also matches provider names as well as course titles and codes.

Filter	Type	Meaning
`keyword`	text	Words in the course title, provider name or a course code
`provider_name`	text	Part of the provider's name
`provider_cricos_codes`	list	CRICOS provider codes, for example `["00025B"]`
`provider_ids`	list	StudySearch provider ids (`provider:cricos:00025b`)
`country_codes`	list	`AU`, `NZ`, `CA`
`subdivision_codes`	list	States and regions, for example `AU-VIC`
`cities`, `postcodes`	list	Campus city (or metro area) and postcode
`regional_categories`	list	1 major city, 2 regional city, 3 regional centre or other
`study_level_codes`, `study_area_codes`	list	From `reference`
`institution_types`, `university_groups`	list	From `reference`
`delivery_modes`	list	`on_campus`, `online`, `blended`, `on_campus_and_online`
`intake_months`	list of 1–12	Courses with an intake in any of these months
`english_test_codes`	list	`IELTS`, `PTE`, `TOEFL_IBT`, `CAE`, `DET` (Duolingo, overall 10–160)
`english_score_max`	number	The student's overall score: courses whose overall requirement is at or below it (needs exactly one test)
`english_band_max`	number	The student's lowest band: courses whose every band requirement is at or below it (needs exactly one test)
`tuition_annual_min`, `tuition_annual_max`	number	Annual tuition range in the course's currency
`has_intake`, `has_english`, `has_tuition`, `has_official_url`, `has_scholarship`	true	Only courses that have this
`changed_since`	date-time	Courses changed since then

`sort`: `relevance` (default with a keyword), `title_asc` (default without), `tuition_annual_asc`, `tuition_annual_desc`, `duration_asc`, `updated_asc`. `page_size` is 1–50.

Example: Nursing in Australia, IELTS 7.0 overall with no band under 6.5, intake in February or July (118 courses on 19 October):

```
{ "action": "search", "filters": { "keyword": "nursing", "country_codes": ["AU"],
"english_test_codes": ["IELTS"],
  "english_score_max": 7, "english_band_max": 6.5, "intake_months": [2, 7] }, "page": 1,
"page_size": 10 }
```

Course detail: `{"action": "course", "cricos_course_code": "082960F"}` (or `course_id`). Several at once (up to 50, in the order asked; ones not found are listed in `not_found`): `{"action": "courses", "identifiers": ["082960F", "0100879"]}`.

Recommended cache: `reference` once a day (it holds countries, states, cities, study levels and areas, institution types, university groups, ranking systems and scholarship vocabularies, with counts). Do not cache search results for more than 15 minutes.

How the screen works in Creator

1. The counsellor fills the search form (keyword, provider, state or city, study level and area, intake months, English test with score and lowest band, tuition range). Build the drop-downs from the cached `reference`.
2. On Search, a Deluge function sends one `search` call through the Connection with those filters, `page` and `page_size` (20 is a good default).
3. Show the returned `items` in a page or widget: title, provider, CRICOS code, campuses, intakes, English, tuition, scholarship count. Use `total` for "N courses" and next and previous page buttons (send `page` + 1).
4. On a course, show the detail from the copy (5.3) or call `course` with its `cricos_course_code`.
5. "Report a problem" on a course sends `report` with the counsellor's note.

5.3 A copy in Creator: seed once, then only what changed

Use this for Institutes and Courses (and optionally Campuses and Contacts), whether or not search runs on it.

1. **Forms.** Create forms Institutes, Courses (with a lookup to Institutes), and optionally Campuses and Contacts, with fields named after the CSV columns (section 6.1). Make `cricos_provider_code` and `cricos_course_code` unique. Add a hidden form, Sync State, with one record holding the cursor, the time of the last run and the last `request_id`.
2. **Seed from CSV.** Call `export` for each entity (`provider` first, then `course`, `campus`, `contact`), page by page while `has_more` is true, and import each file into its form with Creator's import (match on the CRICOS code). The course file is nine files of up to 3,000 rows. Save the `sync_cursor` from the first file you export into Sync State.
3. **Keep current with `sync`.** A scheduled Deluge function, every 15 to 60 minutes: read the cursor; call `sync` with it and `page_size` 50; for each item, `upsert` updates the record with that key or adds it, `remove` marks it inactive (do not delete; a course can come back); save `next_cursor` in Sync State only after the page is written; repeat while `has_more` is true and the run has statement budget left (the next run carries on from the saved cursor).
4. **Map fields.** A `sync` course item holds more than the CSV. Fill the same Creator fields from it: lists (states, campus cities, intake months, delivery modes) joined with `;` as in the CSV; `ielts_overall`, `ielts_band` and `pte_overall` from `english_requirements` (the entry with that `test_code`: `min_overall_score`, `min_band_score`); `tuition_annual` from `tuition.annual_amount`. Add more columns (for example TOEFL or DET) the same way if you search on them.
5. **Searching the copy.** Creator report filters, or a Deluge fetch with criteria, on the stored fields: for example IELTS overall at or below 7.0 and lowest band at or below 6.5 (`ielts_overall <= 7.0 && ielts_band <= 6.5`). Keep intake months in a multi-select field (1 to 12) rather than text, so a filter for February does not also match December ("12"). A course with no IELTS value is "not held", not "no requirement": leave it out of an English filter rather than treat it as a pass.

Sizes to plan for. Since the first seed about 800–950 items have changed a day (mostly courses), so a delta run usually takes one or two calls. A `sync` page holds up to 200 size units (about 600 KB, well under the 5 MB a Deluge `invokeUrl` answer may hold). Deluge allows 5,000 statements in one function run, so write a page of 50 items per call rather than 200, and let a long catch-up continue over several runs. Check your Creator plan's allowances for records, scheduled runs a month and `invokeUrl` calls a day before choosing B for search: each live search (A) is one `invokeUrl` call.

6. Zoho CRM: Institutes and Courses kept current

Australia is in the CRM feed now; New Zealand and Canada can be switched on later without a change to your side (the `reference` answer lists the countries under `sync.countries`).

CRM module	Entity	Key to match on	Count (19 Oct)
Institutes	`provider`	`cricos_provider_code` (also `provider_id`)	1,546
Courses	`course`	`cricos_course_code` (also `course_id`); link to the institute by `provider.cricos_provider_code`	26,103
Campuses (optional)	`campus`	`campus_id`; link by `cricos_provider_code`	3,957
Contacts (optional)	`contact`	`contact_id`; link by `cricos_provider_code`	2,794

Contacts are of four types: `head_of_institution` (the principal executive officer from the public CRICOS register, 1,537), `general` (the international or general enquiry contact from the provider's own website, 1,230), and `team` and `staff` (international recruitment contacts held by StudySearch, 27).

6.1 First import: CSV

`{"action": "export", "entity": "provider"}` returns a CSV file (`text/csv`, header first, one row per item). Entities: `provider`, `campus`, `contact`, `course`. Add `"format": "json"` to get the CSV inside JSON instead.

Files come in pages of up to 3,000 rows. Each page is a complete CSV file with the header. Ask for `"page": 2`, 3 and so on while `x-has-more` is `true`; call one page at a time. On 16 October the course file is nine pages and the other files are one or two.

Header (JSON field)	Meaning
`x-rows` (`rows`)	Rows in this page
`x-page`, `x-pages` (`page`, `pages`)	This page and how many there are
`x-total-rows` (`total_rows`)	Rows in the whole file
`x-has-more` (`has_more`)	`true` while there is another page
`x-complete` (`complete`)	`true` once the platform has built every item of that entity
`x-sync-cursor` (`sync_cursor`)	Where `sync` continues from: keep the one from the first page you export

File	Columns
provider	provider_id, cricos_provider_code, name, legal_name, institution_type, country_code, state, city, postcode, website, phone, email, course_count, campus_count, contact_count, qs_rank, the_rank, updated_at
course	course_id, cricos_course_code, title, provider_id, cricos_provider_code, provider_name, study_level, study_area, country_code, states, campus_cities, delivery_modes, duration_months, intake_months, ielts_overall, ielts_band, pte_overall, tuition_annual, tuition_currency, tuition_year, official_url, scholarship_count
campus	campus_id, provider_id, cricos_provider_code, provider_name, name, address_line1, address_line2, city, state, postcode, country_code, latitude, longitude, metro_area, regional_category, course_count
contact	contact_id, provider_id, cricos_provider_code, provider_name, type, name, title, team, email, phone,

File	Columns
	source, source_url, checked_at

Lists inside a cell are separated by `;`. Import Institutes first, then Courses (so the institute lookup resolves), then Campuses and Contacts.

6.2 Staying current: sync

`sync` returns items in order, each with `entity`, `id`, `seq`, `change` (`upsert` with the full `item`, or `remove`) and a `next_cursor`.

1. **Full load (only if you did not seed from CSV):** `{"action": "sync", "full_load": true}`. Repeat with `{"action": "sync", "cursor": "<next_cursor>"}` while `has_more` is true. About 34,400 items for Australia (about 76 MB in about 200 pages); a page holds up to `page_size` size units (at most 200, about 600 KB), so a page of large items holds fewer. Always follow `has_more`, never count items.
2. **After a CSV seed:** use the `sync_cursor` from the first page you exported as your cursor. Changes made after that file was read then come through `sync` (some may repeat what a later file already holds; upserts by key make that harmless).
3. **Every 15–60 minutes:** send the stored cursor; apply each upsert by key and each remove (mark the record inactive in Zoho rather than deleting it); store the new `next_cursor` only after the page is written. Repeat while `has_more`. Since the first seed about 800–950 items have changed a day, so a run usually takes one or two calls.
4. `entities` limits a full load to some of `provider`, `campus`, `contact`, `course`.

What we expect from the sync job: one cursor at a time (never two runs in parallel), honour `Retry-After`, write pages in bulk, and keep the cursor in Zoho so a failed run resumes where it stopped.

7. Actions

Action	Send	Returns
`reference`	—	Vocabularies with counts; `sync` (entities, countries, item counts)
`search`	`filters`, `page`, `page_size`, `sort`	Courses (section 5.2)
`course`	`cricos_course_code` or `course_id`	One course in full
`courses`	`identifiers` (up to 50)	Those courses, and `not_found`
`providers`	`filters` (`provider_name`, `provider_cricos_codes`, `country_codes`, `subdivision_codes`, `cities`, `institution_types`, `university_groups`, `has_logo`, ranking filters), `page`, `page_size`, `sort`	Providers with courses
`provider`	`cricos_provider_code` or `provider_id`	One provider with its campuses, contacts, rankings (with indicators) and QILT outcomes
`campuses`	`filters` (`provider_cricos_codes`, `provider_ids`, `country_codes`, `subdivision_codes`), `page`, `page_size`	Campuses with address, location and course count
`contacts`	`filters` (as campuses, plus `contact_types`), `page`, `page_size`	Contacts
`statistics`	`kind` `qilt` (filters	QILT measures per provider; PRISMS

Action	Send	Returns
	`provider_cricos_codes`, `provider_ids`) or `prisms` (filters `subdivision_codes`, `sectors`), `page`, `page_size`	international enrolments per region, sector and study area
`rankings`	`filters` (`system_codes`, `edition_year`, `country_codes`, `rank_max`)	Ranking entries with attribution
`scholarships`	`filters` (`student_types`, `open_only`, `course_id`, `nationality_codes` and more)	Scholarships
`scholarship`	`scholarship_id`	One scholarship
`sync`	`cursor`, `full_load`, `entities`, `page_size`	Changes (section 6.2)
`export`	`entity`, `page`, optional `format: "json"`	One page of the CSV file and the sync cursor to continue from (section 6.1)
`report`	`cricos_course_code` or `course_id`, `field`, `note`, `reporter` (`name`, `email`, `ref`)	Records a counsellor's report of a value that looks wrong; StudySearch reviews it

8. Data notes

- **Tuition.** `tuition` is the provider's current fee for international students: the latest year held (`year`), its `basis` and `annual_amount` (worked out from a whole-course fee where only that is published, with `annual_is_derived: true`). The registered total course fee from CRICOS is separate and never merged into it.
- **English.** Each requirement has `test_code`, `min_overall_score` and `min_band_score` (the lowest band asked). An empty list means not held, not "no requirement".
- **Statistics grain.** QILT measures belong to the provider (and student level); PRISMS counts belong to a region, sector and study area. Neither is a course fact: do not show them as one.
- **Scholarships.** `scholarship_count` and the scholarship list are what StudySearch has matched to the course; an empty list does not prove there is none.
- **Logos.** `logo.url` is a signed link valid for a year (`url_expires_at`). Copy the image into Zoho if you keep it; providers are sent again in `sync` before their link ends.
- **Canada** has fewer fields held than Australia and New Zealand. Show what is held.

9. What is held now (19 October 2026)

	Australia
Providers / courses / campuses / contacts	1,546 / 26,103 / 3,957 / 2,794
Courses with intakes	9,977
Courses with English requirements	16,788 (427 accept DET)
Courses with tuition	25,720
Courses with a matched scholarship	9,397
Courses with an official course link	18,593
QILT measures (all providers) / PRISMS rows	3,068 / 2,270

Coverage grows as data is admitted; the CSV and `sync` carry it to you.

10. Acceptance tests

#	Send	Expect
1	`{"action":"reference"}`	200; `sync.countries` is `["AU"]`
2	Section 5.2 example	200; `total` about 118; every item has `cricos_course_code` and `english_requirements`
3	`{"action":"search","filters":{"provider_name":"Queensland"},"page_size":2}`	200; two items, `total` above 1,000 (1,068 on 19 October)
4	`{"action":"course","cricos_course_code":"082960F"}`	200; Bachelor of Nursing (Honours), The University of Queensland
5	`{"action":"courses","identifiers":["082960F","0100879","NOPE"]}`	200; two items; `not_found` is `["NOPE"]`
6	`{"action":"provider","cricos_provider_code":"00025B"}`	200; campuses and contacts listed
7	`{"action":"statistics","kind":"qilt","filters":{"provider_cricos_codes":["00025B"]}}`	200; `total` 48
8	`{"action":"export","entity":"provider"}`	200; CSV with header and 1,546 rows; `x-pages` 1; `x-sync-cursor` present
9	`{"action":"sync","full_load":true,"page_size":5}`	200; at least one item (up to five) and a `next_cursor`
10	`{"action":"search","page_size":500}`	400 `INVALID_INPUT`
11	`{"action":"export","entity":"course","page":9}`	200; 2,103 rows; `x-has-more` false
12	A wrong key	401 `AUTHENTICATION_REQUIRED`

11. Moving from v1

The v1 address (`zoho-course-api`) and its key keep working unchanged until you have moved. StudySearch will agree the switch-off date with you.

v1	v2
`lookup` with `identifier`	`course` with `cricos_course_code` or `course_id`
`search` with top-level `query`, `country_codes`, `limit`, `offset`	`search` with `filters.keyword`, `filters.country_codes`, `page`, `page_size`
`has_intake`, `has_english` flags only in results	Intakes and English requirements in every result
`provider_options`	`providers` (with `filters.provider_name`)
`filter_options`, `reference_bundle`	`reference`
`scholarships` with `course_id`	`scholarships` with `filters.course_id`
`report`	`report` (same fields)
QILT and PRISMS “not admitted”	`provider` (QILT outcomes) and `statistics`
`publication_status` “unpublished”	Every returned course is `published`

12. Support

Send the action, the JSON body (without the key), the HTTP status, the full response and the `request_id` to the StudySearch Platform Admin.

13. Changes to this guide

Issue	Date	API change
First issue	16 Oct 2026	Zoho API v2 at its own address with its own keys: fifteen actions; search results with intakes, English (overall and bands) and tuition; provider name and CRICOS filters; CRICOS codes on every Australian course and provider; campuses, contacts and statistics; CSV export in pages (with the cursor to continue from) and sync for the CRM (Australia)
Issue 2	19 Oct 2026	New English test code DET (Duolingo English Test, overall score 10–160) in `english_requirements`, the `english_test_codes` filter and `reference`